Physics 50 Module 2 Oral Report Grading Rubric

For Module 2, the oral report is ten minutes total: a **two-minute** lightning talk from one slide, then eight minutes of discussion with your instructor. You should bring your laptop and have your slide, Google Doc lab notebook and Google Data Sheet open and ready before you begin.

We will partially assess your skill in communicating your results, but we are not expecting a memorized and polished talk. We are primarily interested in your command of the principles you applied in obtaining your results. We are not testing your recall of the lab manual, but rather, we want to understand the decisions you made in your data collection, analysis, and interpretation.

Assessment Criteria

Your grade will be determined from your performance in each of these approximately equally weighted criteria. Most of your grade is determined by the discussion rather than by the two-minute lightning talk.

1. Two-minute talk
2. Procedure (assessed based on the talk and discussion)
3. Analysis (assessed based on the discussion)
4. Uncertainties and error analysis (assessed based on the discussion)
5. Conclusions and interpretation (assessed based on the talk and discussion)

Every criterion is scored in one of four bins. We are evaluating performance, not counting errors. This means that a single slip up does not drop you a level and a strong answer in one place does not necessarily lift you up one.

- **Strong** : Went beyond expectations, did what a practicing scientist would do.
- **Solid** : Met expectations, with potentially minor soft spot(s) or needing a prompt.
- **Developing** : The pieces are there but not connected or not owned.
- **Not yet** : Not demonstrated.

Logistics

- **Slide due:** Put your slide into your Physics 50 Google Drive folder before your meeting.
- **Materials:** Slide, Google Doc and Google Sheet open, on screen, before you start. Time spent hunting for a file comes out of the discussion time.
- **Presenting:** You should practice the two-minute talk for time; we will cut you off at 2 minutes. During the discussion, we will direct questions to you from any category.
- **Recording**: Your instructor might record the report for their later reference.

Notes specific to Module 2

- In Module 2, you had three quantitative results - the diffraction grating spacing and the number of pixels along both directions of the LCD panel. **Present your final values for all three results, and choose one to support with a plot and diagram(s) during your two-minute talk**. Have plots and diagrams for other results easily accessible (on backup slides or easy to find in your data sheet / lab notebook) and be ready to answer questions about all of them.
- Your slide should include a graph demonstrating a systematically varied quantity, fitted in the Phys 50 plotter, and you should expect questions on the fit and its statistics.

Criteria Expectations

1. Two-minute talk

One slide, with the presentation finished *within two minutes* (your instructor will have a timer set). The title states the result as a claim, not a topic. Opens with the specific question the measurement was meant to answer. An “Approach” schematic and a “Result” figure, both large and readable. Axes on graphs should be labeled with units; the number of trials and what the error bars represent are stated on the slide. Every quantity is quoted with an uncertainty and sensible significant figures. Describe out loud what is plotted before saying what it means.

2. Procedure

The approach schematic makes it obvious what was varied and what was held fixed. Explains the setup well enough for someone with the equipment to repeat it, and states clearly what counted as one trial. Provides a clear justification for the design choices on request (e.g. how many trials, which values, in what order, and why).

No credit here unless procedure and data were documented in Google Doc/Data Sheet during the lab period.

Sample discussion questions about "Procedure" (not a complete list):

- What exactly counted as one trial? Why that many trials? What did you "reset" in between trials? Why?
- Why those values, and in that order? What did you deliberately hold fixed, and how did you check that it stayed fixed?
- What sequence of iterations led to your final experiment? Is this narrative captured in your lab notebook?

3. Analysis

The student moves around their own spreadsheet without hesitation, finds the cell we ask about, and explains in words what the formula does and why that is the right operation. Traces the full chain from a raw reading to the reported value, and says how that value was arrived at — single measurement, mean over N trials, something else. Explanations hold up when follow up questions are asked. The report includes plot(s) systematically varying the quantity on the x-axis over several values. This data is correctly fit in the Phys 50 plotter and the student is able to identify observed trends in fits and fit statistics.

No credit here unless the calculations were applied in the Google Sheet.

Sample discussion questions about "Analysis" (not a complete list):

- You said the error bar was propagated from the SEM of your distance measurement — show me in your spreadsheet how you calculated your error bar, and talk me through the calculation.
- Which cell holds your final answer? Walk me from one raw reading to that cell.
- Why does that step divide by that factor? Where does it come from?

4. Uncertainties and error analysis

The student states exactly what the quoted uncertainty represents (e.g. standard deviation over N trials, SEM, propagated from a particular measurement, reading resolution), shows the propagation in the data sheet, and says where the formula came from. Can distinguish random from systematic error and can explain which error dominates. Reasons correctly about hypotheticals: what changes if N doubles, and what does not.

No credit here unless the uncertainty calculations were applied in the Google Sheet.

Sample discussion questions about "Uncertainties and error analysis" (not a complete list):

- Is that a standard deviation or a standard error of the mean (SEM)? Which one belongs here, and why?
- If you took twice as many trials, what do you expect to happen to your reported value and to its uncertainty? How would that change your interpretation?
- Which measurement contributes more to your final uncertainty? How do you know?
- Is there an error here that more trials would not fix?

5. Conclusions and interpretation

The conclusion answers the question that opened the talk, and every result is quoted with its uncertainty. Weighs the size of the effect against that uncertainty. Names the dominant source of error and proposes a specific change that would reduce that source. Keeps observation separate from interpretation and does not overinterpret.

Sample discussion questions about "Conclusions and interpretation" (not a complete list):

- You measured $x \pm \delta x$. What conclusions can you draw? Is that a real effect? How do you know?
- What would you change first if you had more time or were to do it again, and what specifically would it fix?